iGEM PATRAS · DRY-LAB / MODELLING

Enkephalin Analgesia by PENK Expression + NEP Knockdown

A four-workstream computational audit & design package

Plasmid audit · Restriction QC · siRNA design · Lipoplex CG-MD

The therapeutic idea

Make more enkephalin · degrade less of it · at the pain relay

- **Make more:** a **PENK** plasmid expresses proenkephalin → processed to enkephalins → endogenous opioids acting on δ/μ receptors.
- **Degrade less:** silence **neprilysin (NEP/MME)**, the enzyme that inactivates enkephalin in seconds — an **siRNA** knockdown arm.
- **Deliver both together:** co-formulated in a **lipoplex**, delivered locally (intrathecal, spinal dorsal horn).

Opioid-like analgesia without an exogenous opioid. Clinical precedent exists for enkephalin gene therapy in pain.

1 · Plasmid audit — one real defect found

#	Plasmid	PENK	Verdict
1	mEGFP-IRES-PENK	**TRUNCATED (6/7 cores)**	**DEFECT**
2	mEGFP-P2A-PENK	full length	OK, secretion caveat
3	PENK-IRES-mEGFP	full length	**USE THIS — cleanest**
4	PENK-P2A-mEGFP	full length	OK (backup)

- **Use plasmid 3** — PENK cap-driven, signal peptide intact, byte-identical to reference.
- **Plasmid 1's truncation removes the 7th enkephalin core — and is invisible on a gel. Only Sanger sequencing catches it.**

2 · Restriction-digest QC — before every transfection

Digest A — insert release

AflII + XbaI → 5,345 bp backbone (control) + linker band:

- **~2.16 kb = IRES**
- **~1.63 kb = P2A**

Digest B — identity fingerprint

NcoI (5 cuts) → unique 5-band pattern; second-largest band = gene order.

What it catches

Mis-assembly, wrong linker, wrong gene order — before wasting a transfection.

What it can't

The plasmid-1 point truncation. Gel QC does not replace Sanger sequencing.

Virtual gel + full recipe delivered as protocol + figure.

3 · Anti-NEP (MME) siRNA — knockdown reagents

2,084 sites screened → sequence rules → thermodynamic asymmetry → off-target → **accessibility**. All target the CDS shared by **all 4 MME variants**.

Priority	Name	Guide (5'→3')	Notes
1st	si-MME-3 (CDS1224)	`AUAGUUUGCACAACGUCUC`dTdT	perfect specificity + open site
2nd	si-MME-4 (CDS1590)	`UAUCCACUCAUCUUUGUCC`dTdT	most accessible target
3rd/4th	si-MME-2 / si-MME-1	CDS1032 / CDS759	0 off-targets ≥17/19

- Validate ≥2 by qPCR at 48 h (≥70% knockdown) + CD10 Western; scrambled control mandatory.

4 · Lipoplex CG-MD — does pH control DNA binding?

MARTINI coarse-grained MD, 3 seeds × 3 formulations. Question: does the ionizable lipid bind DNA at endosomal pH but release at physiological pH?

Formulation	DNA–lipid contacts (50 ns)	Reading
DOTAP (permanent +)	21.2 ± 6.1	binds constitutively
MC3-H$^+$ (endosomal pH)	32.5 ± 15.1	**binds strongly**
MC3^0 (physiological pH)	9.6 ± 5.2	does not reliably bind

- **The pH switch is reproduced** → favours an ionizable lipid over permanent-cation.
- Large error bars = binding still nucleating at 50 ns → **200 ns extension running** (honest convergence need, not padding).

Why co-formulation is non-negotiable

Silencing is cell-autonomous — but enkephalin degradation is not.

- NEP degrades enkephalin **extracellularly**, on the surface of **many** cells.
- siRNA only silences the cells it actually enters.
- ⇒ Silencing NEP only in the enkephalin-producing cell is **insufficient** — neighbours still degrade the secreted peptide.
- **⇒ Co-package siRNA + PENK plasmid in the same lipoplex, delivered to the same locus. This is what makes the lipoplex load-bearing.**

Honest limitations & the one-line plan

Stated by design, not hidden:

- MD lipid topologies are MARTINI-2 **analogy models** → conclusions are **qualitative/comparative**, not binding free energies.
- Cell-autonomy mismatch is the central design risk (mitigated, not erased, by co-formulation).
- NEP has other substrates (Aβ, substance P) → knockdown kept **local & transient**.

Use plasmid 3 → QC by gel + Sanger → order si-MME-3 (+1 more)
→ co-formulate in an ionizable-lipid lipoplex → deliver intrathecally.